Chirayu Nilesh Chaudhari

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Summary

Driven second-year Artificial Intelligence & Data Science student with a strong foundation in **full-stack development**, **artificial intelligence**, and **hardware integration**. Proven ability to lead and collaborate on technical projects and campus events. Eager to contribute my diverse skills to help solve complex, real-world challenges.

Skills

Python, JavaScript, React, Node, Next, Java, AI/ML, Data Structures & Algorithms, IoT/ Embedded Systems, Signal Processing

Publications & Honors

Conferences Published:

- Chirayu Nilesh Chaudhari, et al. 'Contour based tracking vs Hough transform based tracking: A RC Car experiment.' *3rd IEEE World Conference on Communication & Computing (WCONF-2025)*, Kalinga University, 2025.
- Chirayu Nilesh Chaudhari, et al. 'The Effects and Classification of Psychoactive Substances: A Machine Learning Approach.' Accepted for presentation at the *IEEE International Conference on Computational Systems and Information Technology for Sustainable Solutions (CSITSS 2025).*

Journals Published:

- Chirayu Nilesh Chaudhari, et al. 'Thermoelectric Energy Optimization with Deep Reinforcement Learning.' Accepted for publication in a *Springer Nature* journal.

Achievements:

- **ICPC Regionalist (2025):**
  Qualified to compete in the International Collegiate Programming Contest (ICPC) Regionals, showcasing top-tier problem-solving and algorithmic optimization skills under time pressure.
- **National Finalist (Top 6) – Novo Nordisk Hackathon (Sep 2025 – Oct 2025)**
  Placed top 6 nationally in a highly competitive hachathon focused on healthcare innovation.

Experience

AI Engineering Intern Dec 2025 – March 2026
ApplyBeeAI, Bengaluru, Karnataka

- Engineered core NLP and agentic capabilities using LangChain, spaCy, and Gemini, scaling automated processing pipelines across 7+ applicant tracking systems, including Naukri and Workday.
- Designed, containerized with Docker, and deployed core AI product modules to GCP, collaborating with founders to turn complex requirements into functional, highly scalable early-stage solutions.

Intern May 2025 – Jul 2025
SigNoz, Bengaluru, Karnataka

- Architected and developed a Model Context Protocol (MCP) server from scratch using Node.js and TypeScript to streamline data access for AI-powered clients.
- Integrated official SigNoz APIs to fetch and process real-time trace and log data, enabling comprehensive observability.
- The server was engineered to ensure seamless accessibility from various GPT-based clients, including Cursor, Windsurf, and Claude.
- Conducted research on existing solutions, such as Honeycomb's MCP, to improve the server's design and implementation strategy.

Projects

Suspicious Signal Detection using ML & SDR Nov 2025 – Dec 2025
Engineered a dual-layer anomaly detection pipeline using ADALM-PLUTO for RF I/Q capture in ISM bands. Developed a synthetic data generator to inject QPSK bursts and benchmarked metadata-based classifiers (SVM RBF, Random Forest) against a custom PyTorch 1D CNN trained on raw I/Q samples, demonstrating superior detection of temporal and spectral anomalies.

Facial Emotion Recognition Experiments on KDEF Oct 2025 – Nov 2025
Built a reusable pipeline that contrasts HOG+LinearSVC baselines with fine-tuned CNN backbones on the KDEF dataset, packaged shared `fer` utilities for loaders, feature builders, and training controls, and automated metrics, reports, plots, and model exports with flexible CLI toggles.

Thermoelectric Energy Optimization with Deep RL Aug 2025 – Sep 2025
Developed a Gymnasium-compatible thermoelectric generator environment and orchestrated Optuna-tuned PPO, A2C, DDPG, SAC, and DQN agents in Stable-Baselines3, producing benchmark comparisons, evaluation plots, and reusable training utilities for energy management research.

LLM Optimized Synthesis of Gold Nanoparticles Jan 2025 – Jul 2025
Engineered a novel system using a Large Language Model (LLM) via the Gemini API and a Keras deep learning model to optimize the synthesis of Gold Nanoparticles (AuNPs) for cancer treatment. The system predicts AuNP properties from synthesis parameters and leverages the LLM to analyze results and provide intelligent suggestions for process improvement.

Energy Conservation with Custom LWT Jan 2025 – Jul 2025
Implemented a custom Lifting Wavelet Transform (LWT) from scratch for data preprocessing, which was then used in a custom Neural Network for energy consumption analysis. Also developed a UI for users to apply the LWT to their own N-D datasets.

The Effects and Classification of Psychoactive Substances: A Machine Learning Approach Sep 2024 – Jun 2025
Developed a multi-stage machine learning application with a Flask front-end to predict psychoactive substance use from personality traits, forecast behavioral traits, and classify substances based on chemical properties.

Education

Bachelor's of Technology in Artificial Intelligence & Data Science Aug 2024 – May 2028
CGPA: 9.33
Amrita Vishwa Vidyapeetham, Bengaluru, Karnataka